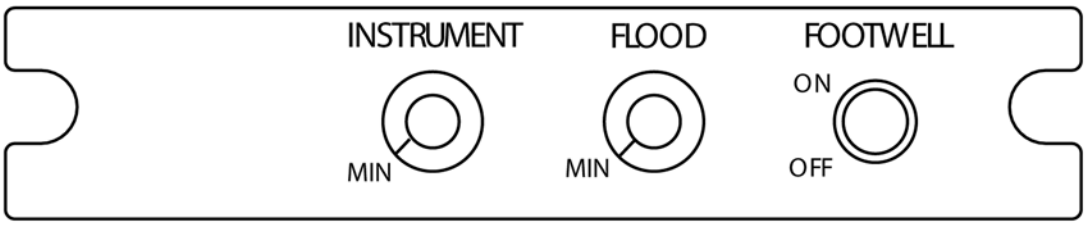
Instrument Panel Lighting System – General data

1. General

Instrument Panel Lighting system provides lighting to the instrument panel. The system consists of the dimming rheostat, the Lighting Power Supply (LPS), and the instrument panel lights. The instrument dimming rheostat is located on the dimming control panel mounted forward of the multi-function controller. Refer to Figure 1. The individual instrument panel lights are powered by the LPS through dedicated outputs. The output voltages of the LPS to the instrument lights are adjusted by the rheostat.



Control Column Dimming Panel
Figure 1

2. Description

A. Lighting Power Supply (LPS)

The LPS supplies 5V and 28V variable voltage to panel lights, avionics instruments, static valve, and the standby compass lighting. The dimming is controlled by a 10k Ω rheostat on the Control Column Dimming Panel. The LPS supplies 0-5 volt variable voltage to the Fuel Selector Panel overlay, Overhead Switch Panel overlay, Left Hand Panel overlay, Right Hand Panel overlay, Cockpit Dimmer Panel overlay, and TB13. The LPS supplies 0-28V variable voltage to the avionics instruments, standby compass, TB13, trim panel overlay, flap selector post lights, and the (ESIS) Alternate Static Valve. The LPS also includes a 9-pin maintenance port, which is utilized by any PC computer equipped with RS422 adapter unit maintenance or set up. The display revisionary panel is supplied by the 0-28V output through R101, a 150 Ω drop resistor, so as to supply 14V max to the panels.

The LPS output voltages are controlled by the instrument dimming rheostat. The lights that are illuminated and dimmed through the LPS are listed in Table 1 — LPS Lighting Interconnect. The LPS is located on the lower left hand tray of the forward avionics bay. The unit is accessible from the forward baggage compartment by removing the lower HIRF protection cover.

28 Volt DC power is protected by a 5 Amp PANEL LIGHTS L (P7) (Left Bus) circuit breaker located on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

Table 1 LPS Lighting Interconnect

LPS Pin	Signal Name	LPS Table Ref.	System (DHC–6–400) Mod 6/2028
1	Output 1 (0-28V)	Table 16	HF (S.O.O. 6215)
2	Output 2 (0-28V)	Table 17	Audio Panel, R/H
3	Output 3 (0-28V)	Table 18	Audio Panel, L/H
4	Output 4 (0-28V)	Table 19	FD/AP (S.O.O. 6221)
5	Output 5 (0-28V)	Table 20	FD/AP (S.O.O. 6221)

Table 1 LPS Lighting Interconnect (continued)

LPS Pin	Signal Name	LPS Table Ref.	System (DHC–6–400) Mod 6/2028
6	Output 6 (0-28V)	Table 21	Reversionary Panel
7	Output 7 (0-28V)	Table 22	Apex Panels
8	Output 8 (0-28V)	Table 23	Apex Panels
9	Output 9 (0-28V)	Table 24	Apex Panels
10	Output 10 (0-28V)	Table 25	PFD, L/H
11	Output 11 (0-28V)	Table 26	PFD, R/H
12	Output 12 (0-28V)	Table 27	PFD, Upper
13	Output 13 (0-28V)	Table 28	PFD, Lower
14	Output 14 (0-28V)	Table 29	ESIS (Except C-FDHT)
15	Output 15 (0-28V)	Table 30	Clock
16	Output 16 (0-28V)	Table 31	Trim Panel, Flap Sel, Alt Static
17	Output 17 (0-28V)	Table 32	Compass
18	Output 18 (0-28V)	Table 33	not used
19	Output 19 (0-28V)	Table 34	not used
20	Output 20 (0-28V)	Table 35	not used
21	Output 21 (0-28V)	Table 36	not used
22	Output 22 (0-28V)	Table 37	not used
23	Output 23 (0-28V)	Table 38	not used
24	Output 24 (0-28V)	Table 39	not used
25	Output 25 (0-28V)	Table 40	Spare 28 V
52	Output 15 (0-5V)	Table 15	Spare 5 V
53	Output 14 (0-5V)	Table 14	not used
54	Output 13 (0-5V)	Table 13	not used
55	Output 12 (0-5V)	Table 12	not used
56	Output 11 (0-5V)	Table 11	not used
57	Output 10 (0-5V)	Table 10	not used
58	Output 9 (0-5V)	Table 09	not used
59	Output 8 (0-5V)	Table 08	not used
60	Output 7 (0-5V)	Table 07	not used
61	Output 6 (0-5V)	Table 06	not used
62	Output 5 (0-5V)	Table 05	Dim Panel
63	Output 4 (0-5V)	Table 04	Right Switch Panel
64	Output 3 (0-5V)	Table 03	Left Switch Panel

Table 1 LPS Lighting Interconnect (continued)

LPS Pin	Signal Name	LPS Table Ref.	System (DHC–6–400) Mod 6/2028
65	Output 2 (0-5V)	Table 02	Upper Switch Panel
66	Output 1 (0-5V)	Table 01	Fuel Selector Panel

3. Adjustment/Test

A. Function Test Lighting Power Supply (LPS)

1) General

The unit does not require alignments or adjustments. However, if ramp test or maintenance check is required, the unit should be connected to a laptop via the maintenance port located on the maintenance panel in the forward avionics bay. The connection is RS422 type communication. Thus, a RS422 adapter may be used for connection to a laptop. EASY SYNC UNIT P/N ES–U–3001M is recommended for this purpose. Other applicable RS422 to USB adapters may be used as alternatives.

The example setup procedure using the recommended unit is listed below.

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Turn the Instrument dim control from the counterclockwise (MIN) position, to the fully clockwise position on the Control Column Dimming Panel. Verify that the instrument panel lights go from dark to full brightness.

d) Turn the Instrument dim control from the counterclockwise (MIN) position, to the fully clockwise position on the Reversion Control Panel. Verify that the instrument panel lights go from dark to full brightness.

e) Set the Instrument dim control from the full brightness position to a comfortable viewing dim level.

f) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Fault Log Data Test

a) Assemble the RS422 adapter to a RS422 connector (M24308/4–259 preferred) as per PSM 1-64-2W, Chapter 33-10-10.

b) Acquire a USB type B to USB type A adaptor cable (regular commercial printer cable).

c) Acquire a computer equipped with USB port and install applicable RS422 virtual COM port driver based on the computer operating system.

d) Acquire a computer equipped with USB port and install applicable RS422 virtual COM port driver based on the computer operating system.

e) Connect USB type B port to the ES–U–3001M USB port.

f) Connect M24308/4–259 connector to LPS maintenance port.

g) Setup the communication via Windows Hyperterminal

1 Create a new connection in Windows Hyperterminal. Refer to Figure 2.

2 Confirm correct COM selected for USB adapter. Refer to Figure 3.

3 Set the communication protocol as 9600 bits per second, 8 data bits, none parity, 1 stop bit, and none flow control. Refer to Figure 4.

4 The Emulation should be set as Auto detect. Refer to Figure 5.

5 The ASC II should be set to send line ends with line feeds and echo typed characters locally. Refer to Figure 6.

h) Check if the connection is successfully established. Upon establishment of the connection, carriage return should show similar output from the LPS unit. Refer to Figure 7. The recommended RS422 adaptor ES–U–3001M may be ordered from Viking Air Ltd. The LPS unit must be powered for successful connection.

i) Once the connection has been successfully established, there are two commands that may be used for verify the correct operation of the unit. All commands entered into the Hyperterminal should be using UPPER CASE LETTERS ONLY.

j) Type DISP in the command window will output the current operating parameters of the PS 1 unit, including temperature switch status, and output voltage levels. Valid range of voltage outputs should be within the range from 00 to FF in hex number format.

k) Type DFLG in the command window will output the fault log data stored in the PS 1 unit internal memory.

l) A correctly functioning unit should not have any fault log output by entering this command.

4. Removal and Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Removal and Installation of the LPS

1) General

These procedures are for the removal and installation of the LPS. The unit is not considered ESD sensitive. However, a high density connector has been used for the power supply outputs. For removal and installation, extra care should be taken for assembling the cable assembly to the LPS unit. The pins may be easily damaged.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open PANEL LIGHTS L (P7) circuit breaker on the avionics circuit breaker panel and fit with safety clip.

c) Gain access to the forward avionics shelves through the forward baggage compartment.

d) Disconnect the electrical connectors from the front of the unit.

e) Loosen the four screws and nuts used to fasten the unit to the mounting bracket.

f) Remove unit from aircraft.

Note Stow connectors as required.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the PANEL LIGHTS (P7) circuit breaker on the avionics circuit breaker panel is opened and fitted with safety clip.

c) Slide the LPS unit into the mounting tray.

d) Tighten the four screws and nuts used to fasten the unit to the mounting bracket.

e) Reconnect the electrical connectors to the unit. Ensure connections are secure.

f) Remove safety clip and close PANEL LIGHTS L (P7) circuit breaker on the avionics circuit breaker panel.

B. Removal and Installation of the LPS Mounting Tray

1) General

These procedures are for the removal and installation of the LPS Mounting Tray.

2) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove the LPS unit as per Para 4.A.2).

c) Remove screws securing the mounting tray to the instrument panel.

d) Remove mounting tray from aircraft.

3) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Align the tray to the mounting holes.

c) Install the screws that secure the LPS mounting tray to the front avionics rack.

d) Install the LPS unit as per Para 4.A.3).

e) Test the LPS as per Para 3.A.2).

5. Troubleshooting

A. LPS

1) If the LPS is deemed inoperative check the following:

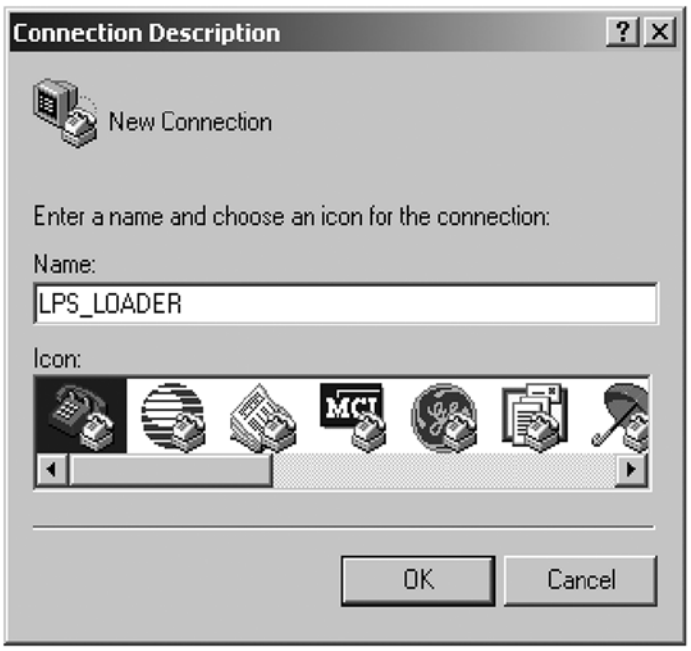
a) Ensure PANEL LIGHTS L (P7) circuit breaker is closed.

b) Check that the units are securely installed and the locking screw is securely tightened.

c) Visually inspect LPS unit connectors for damaged wires or contacts.

d) Check if the unit output voltage is valid at various rheostat locations.

e) If repair or service is required, send to an appropriate rated avionics repair facility.

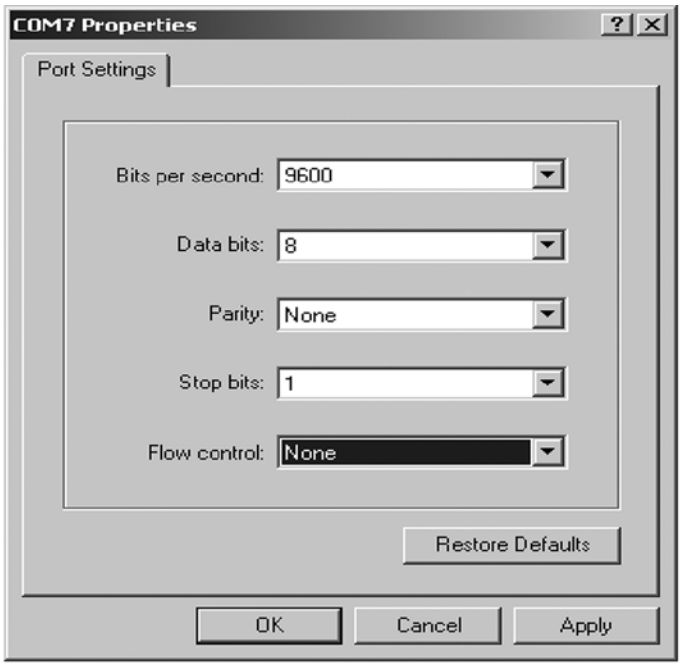


Connection Description
Figure 2

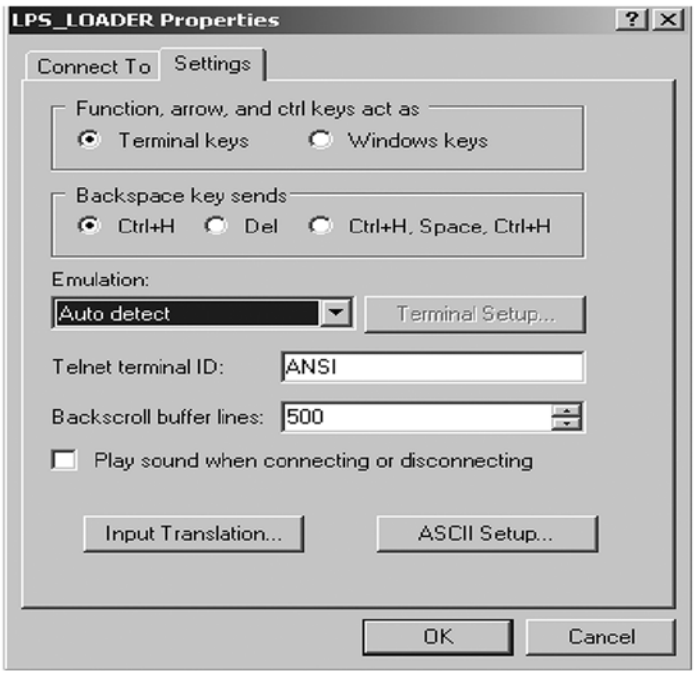


Confirm Correct COM Selected for USB Adaptor
Figure 3

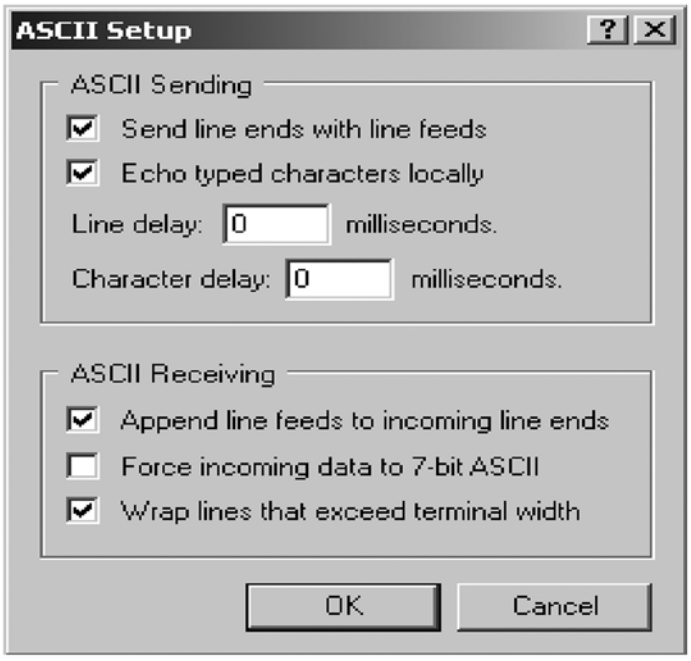
Note COM 7 shown for reference only actual port may be different.



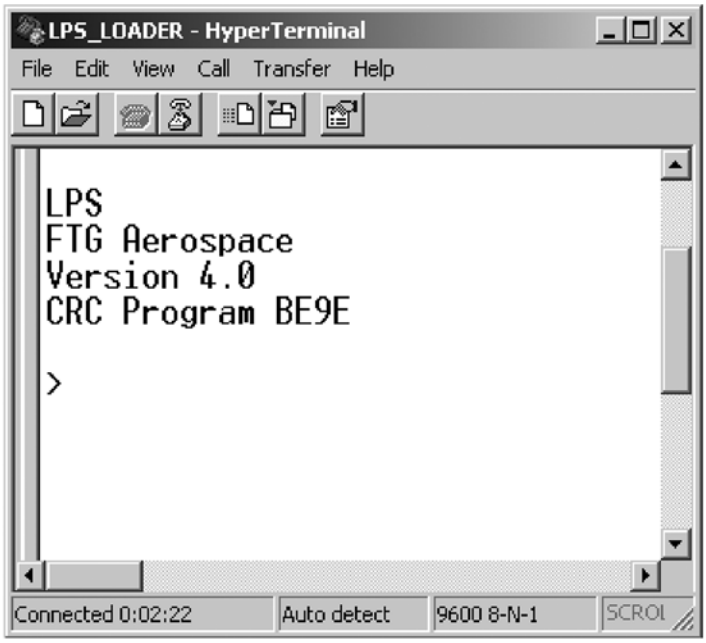
Set the Communication Protocol Window 1
Figure 4



Set the Communication Protocol Window 2
Figure 5



ASCII Setup
Figure 6



Carriage Return Response Upon Successful Connection
Figure 7

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